

Module 5: Information, Feedback, and Cybernetic Systems

Student Summary and Practice

Name: _____

Date: _____

The question for this module. How can a system use information about what just happened to guide what happens next? A thermostat, a driver, a greenhouse controller, and a learning algorithm can all be studied with the same pattern:

$$\text{state} \longrightarrow \text{measurement} \longrightarrow \text{comparison} \longrightarrow \text{action} \longrightarrow \text{new state}.$$

The new state can be measured again, closing the loop. A system organized this way is called a **cybernetic system**.

1. Read the signal

The **state** is the condition that can change. A **measurement** is the information the system receives about that state. A **target** or **reference value** is the condition the system is trying to reach. The difference between the target and the measured value is the **error signal**:

$$e_n = r - x_n,$$

where r is the target and x_n is the measured state at step n . Here, “error” means a difference; it does not necessarily mean a mistake.

Example. A classroom is at 64°F and the thermostat target is 70°F. Then $e = 70 - 64 = 6^\circ\text{F}$. The room is six degrees below its target, so the controller calls for heat.

2. Corrective feedback can be gentle or too strong

A simple correction rule is

$$x_{n+1} = x_n + k(r - x_n),$$

where k is the response strength. It tells the system what fraction or multiple of the current error to correct.

Example. With target $r = 70$, starting state $x_0 = 64$, and $k = 0.5$, the system corrects half of the remaining error each step:

Step n	State x_n	Error $70 - x_n$	Next state x_{n+1}
0	64	6	$64 + 0.5(6) = 67$
1	67	3	$67 + 0.5(3) = 68.5$
2	68.5	1.5	$68.5 + 0.5(1.5) = 69.25$

For this model, $0 < k < 1$ approaches the target without crossing it. If $1 < k < 2$, the system crosses the target and then corrects back; the overshoots shrink. At $k > 2$, the overshoots grow. **Stability** describes whether departures settle down or grow.

3. Feedback can amplify change

Corrective (negative) feedback acts against a departure from a target. The word negative describes the direction of the response, not whether it is harmful. **Amplifying (positive) feedback** reinforces a change, so a change can produce more change in the same direction.

For a simple percentage-growth process,

$$x_{n+1} = (1 + g)x_n,$$

where g is the growth rate. Starting at 100 with $g = 0.20$ gives 100, 120, 144, ... Each increase is calculated from the larger value that came before it.

4. Information is what the system can use

A controller acts on a **signal**, not directly on the world. A signal can be rounded, noisy, incomplete, or delayed. If the controller sees an old measurement, it may correct a condition that has already changed and overshoot.

Information can be measured in bits. For an event with probability p , its surprisal is

$$I = -\log_2(p).$$

An event with probability $1/8$ has $I = -\log_2(1/8) = 3$ bits. A parity bit can help detect some transmission errors, but it does not tell the receiver which bit changed or repair the message. These tools matter because a feedback loop is only as useful as the information it receives.

5. Learning is a feedback process

A prediction by itself is a one-way calculation. Learning closes a loop: make a prediction, compare it with a target, measure the error, change adjustable weights, and predict again. In a real neural network, backpropagation calculates how the final error is connected to weights in earlier layers. A training rule uses that information to update the weights.

attempt $\rightarrow$ error $\rightarrow$ change $\rightarrow$ new attempt

Quick check

Idea	Ask yourself
Feedback loop	What is measured, compared, and changed?
Error	Is the measurement above or below the target?
Response strength	Does the correction approach, cross, or move away from the target?
Signal quality	Is the information current and precise enough for the action?
Information	How many bits correspond to an event with probability p ?
Learning	Does the error from one attempt change the next attempt?

Module 5 Practice

Show your calculations. Then explain what each number means in the system.

1. A greenhouse controller tries to keep the room at 72°F. Its sensor reports 66°F. Identify the target, the measurement, and the error. State what the positive error means.
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2. The target is 100, the current state is 80, and the controller uses $k = 0.25$.

- (a) Find the error.
 - (b) Find the next state using $x_{n+1} = x_n + k(r - x_n)$.
 - (c) How much closer to the target did the system move?
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3. A controller uses the same rule with $k = 1.5$. The target is 100 and the current state is 80.

- (a) Find the next state.
 - (b) Did the system cross the target?
 - (c) In this simple model, what happens to later overshoots when $1 < k < 2$?
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4. A store wants to keep 100 items in stock. Its current inventory is 60, but the controller acts on yesterday's report of 80 and orders half the reported shortage.

- (a) How many items does the controller order from the delayed report?
 - (b) How many would it order if it used the current inventory instead?
 - (c) Explain why delayed information can lead to a poor correction.
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5. A one-weight learning model predicts $\hat{y} = 2w$. The target is 6, and the update rule is $w_{\text{new}} = w + 0.25(6 - \hat{y})$. Begin with $w = 1$.

- (a) Find the prediction and error on the first attempt.
- (b) Find the updated weight.
- (c) Find the next prediction. Did it move closer to the target?